

Miguel Castillo

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EDUCATION

Rutgers University-New Brunswick, NJ

September 2020 - May 2024

Bachelor of Computer Science

- **Skills:** Java, Python, JavaScript, Spring Boot, Django, R, React, HTML, Tailwind CSS, SQL, Zsh, Data Analysis, Excel
- **Languages:** English (Fluent), Spanish (Fluent)
- **Relevant Coursework:** Data Structures (210), Intro to Artificial Intelligence (440), Design and Analysis of Computer Algorithms (344), Principles of Information and Data Management (336), Data Management for Data Science (210)
- **Honors:** Educational Opportunity Fund (EOF) Scholar

WORK EXPERIENCE

System Tester - Analyst Trainee (Full Time)

May 2026 – Present

State of New Jersey - DCF, Information Technology | Trenton, New Jersey

- Performed functional, regression, and User Acceptance Testing (UAT) on NJ SPIRIT, a large-scale enterprise case management platform used across New Jersey child welfare operations.
- Documented and tracked software defects using formal QA workflows, collaborating with developers and business analysts to ensure timely resolution and system stability
- Maintained and updated test scripts, testing documentation, and execution evidence to support audit readiness and system governance standards
- Evaluated system workflows across a complex case management platform, validating business logic, data flow integrity, and application reliability.

Digital Technology Literacy Instructor (Full Time)

November 2025 – April 2026

Catholic Charities of the Archdiocese of Newark | Newark, New Jersey

- Built Google Apps Script (JavaScript-based/.gs) automation to reduce manual administrative workload and improve attendance efficiency.
- Automate data processing, reporting, and classroom workflow integration using Google Workspace APIs.
- Provide technical troubleshooting and system-level support for classroom hardware, software, and networked tools.
- Produce monthly technical training documentation and lesson plans for administrators, introducing automation tools, software workflows, and Excel-based processes.

PROJECTS

Live Interactive Personal Website | *JavaScript, Python, React, Rest APIs, Three JS, Tailwind CSS, Django*

Present

Present – (Tag Manager 700 visits)

- Perceptron & Two Layer Neural Network: Facial recognition and digit classification, achieving over 90% accuracy improvement through optimized data preprocessing and iterative learning.

Flight Travel Reservation System | *Java, Spring Boot, HTML, CSS, SQL, JSP, Git*

October 2023

Rutgers CS 336 - Fall 2023

- Developed a full-stack flight reservation platform using Spring Boot, Java Servlets (JSP) and MySQL, integrating frontend and backend via RESTful APIs for user authentication, flight search, and booking management.
- Conceptualized and crafted an Entity-Relationship (ER) diagram for an insightful database structure, honing data modeling proficiency, secure data validation and session-based authentication.

Grid World Pathfinding Algorithms | *Python, NumPy, Pandas, Matplotlib*

May 2024

Rutgers CS 440 - Spring 2024

- Conducted comprehensive analyses of pathfinding algorithms, including Repeated Forward A* and Adaptive A*, exploring the impact of heuristic consistency and tie-breaking strategies on runtime and cell expansions.
- Leveraged advanced concepts like Manhattan distance consistency and adaptive heuristics to enhance search efficiency while rigorously evaluating performance through detailed experimentation and visualization.

REST-Based React 3D Pokédex Visualizer | *JavaScript, React, Rest APIs, Three JS, Tailwind CSS*

June 2022

Summer 2024

- Developed an interactive Pokédex web application using React (JSX), Three.js, and the Pokémon API, integrating dynamic 3D visualizations for an engaging user experience.